English translation prepared with ChatGPT (OpenAI, GPT-6), 26 October 2026, for translation of the Bulgarian text into English and LaTeX formatting. The Bulgarian originals remain the reference; translation may contain errors.

English transcription of the original handwritten solution notes. Cross-references and the level of detail of the original are retained.

Problems 1 and 2

See Problems 1 and 2 of the Computer Science Test 2, 2024.

Problem 3

See the exercise classes.

Problem 4

Let $X, Y \sim \mathrm{Exp}(1)$ be independent and identically distributed. Recall that $\mathbb{P}(\mathrm{Exp}(\lambda) > t) = e^{-\lambda t}$. For $t > 0$,

$$\begin{aligned}\mathbb{P}\left(\sqrt{X/Y} > t\right) &= \mathbb{P}(X > Yt^2) \\ &= \int_0^\infty \mathbb{P}(X > yt^2) f_Y(y)\,\mathrm{d}y \\ &= \int_0^\infty e^{-yt^2} e^{-y}\,\mathrm{d}y \\ &= \frac{1}{1+t^2}\int_0^\infty e^{-y(1+t^2)}\,\mathrm{d}[y(1+t^2)] = \frac{1}{1+t^2},\end{aligned}$$

using $\int_0^\infty e^{-x}\,\mathrm{d}x = 1$. Therefore,

$$\mathbb{E}\sqrt{X/Y} = \int_0^\infty \mathbb{P}\left(\sqrt{X/Y} > t\right)\,\mathrm{d}t = \int_0^\infty \frac{\mathrm{d}t}{1+t^2} = \frac{\pi}{2}.$$

Alternatively, use $F_{\sqrt{X/Y}}(t) = t^2/(1+t^2)$ and obtain the density by differentiating, $f = F'$, and so on.

Second method

$$\begin{aligned}\mathbb{E}\sqrt{X/Y} &= \int_0^\infty \int_0^\infty \sqrt{x/y}\, f_{X,Y}(x,y)\,\mathrm{d}x\,\mathrm{d}y \\ &= \int_0^\infty \int_0^\infty \sqrt{x/y}\, e^{-x} e^{-y}\,\mathrm{d}x\,\mathrm{d}y \\ &= \left(\int_0^\infty e^{-x}\sqrt{x}\,\mathrm{d}x\right)\left(\int_0^\infty e^{-y}\frac{1}{\sqrt{y}}\,\mathrm{d}y\right) \\ &= \Gamma(3/2)\Gamma(1/2) = \frac{1}{2}\Gamma(1/2)^2,\end{aligned}$$

where $\Gamma(z+1) = z\Gamma(z)$. By the first method,

$$\frac{1}{2}\Gamma(1/2)^2 = \frac{\pi}{2}, \qquad \text{hence} \qquad \Gamma(1/2) = \sqrt{\pi}.$$

Problem 5

See Problem 4 of the Software Engineering examination, February 2024.

Problem 6

This is the proof of a special case of Cochran's theorem, divided into individual steps.

Problem 7

$$e^{-n}\left(\frac{n^0}{0!}+\frac{n^1}{1!}+\cdots+\frac{n^n}{n!}\right)=\sum_{k=0}^{n}\mathbb{P}(\mathrm{Poi}(n)=k)=\mathbb{P}(\mathrm{Poi}(n)\leq n).$$

As seen earlier in the course, if $X_n\sim\mathrm{Poi}(n)$, then

$$X_n\stackrel{d}{=}Y_1+\cdots+Y_n,\qquad Y_1,\ldots,Y_n \text{ i.i.d. } \mathrm{Poi}(1),\qquad \mathbb{E}Y_i=D(Y_i)=1.$$

Thus, by the central limit theorem,

$$\begin{aligned}\mathbb{P}(\mathrm{Poi}(n)\leq n)&=\mathbb{P}(Y_1+\cdots+Y_n\leq n)\\&=\mathbb{P}\left(\frac{Y_1+\cdots+Y_n-n}{\sqrt{n}}\leq\frac{n-n}{\sqrt{n}}\right)\\&\longrightarrow\mathbb{P}(N(0,1)\leq 0)=\frac{1}{2}.\end{aligned}$$